

Memory Timeline

Comprehensive Application Overview & Technology Stack
Specifications

1. EXECUTIVE DESCRIPTION

Memory Timeline is a full-stack personal archive for preserving, organizing, and revisiting image-based memories in a private visual workspace. Users can create rich multi-image entries with dates, categories, descriptions, favorites, and reminders; explore them in timeline, calendar, or compact tile layouts; search, sort, filter, share individual memories, and export image collections. Reminder delivery uses permission-based browser push notifications for each enabled device, backed by a service worker, VAPID, and retry-aware scheduling. The application also provides a PIN-gated Hidden Images interface, 30-day Trash recovery, profile and account controls, responsive layouts, and extensive per-device appearance, sound, and interaction settings.

2. TECHNOLOGY STACK SPECIFICATIONS

Frontend Ecosystem

- **React 19** - Component-driven interface and stateful user workflows
- **Vite 7** - Fast development server and optimized production bundling
- **React Router 7** - Protected routes, public share links, and navigation
- **Framer Motion 12** - Page transitions, onboarding, and shared-layout motion
- **Axios** - API client with credentials, CSRF support, and session refresh
- **Service Worker and Web APIs** - Push handling, Web Audio, Canvas, and local preferences

Backend Ecosystem

- **Node.js and Express 5** - REST API, middleware, and application runtime
- **Mongoose 9** - MongoDB schemas, indexes, queries, and lifecycle state
- **JWT and cookie sessions** - Rotating access and refresh authentication
- **bcryptjs** - Password and Hidden PIN hashing
- **Multer and Sharp** - Validated uploads, WebP normalization, and thumbnails
- **sanitize-html and web-push** - Safe rich text and VAPID-authenticated push delivery

Database, Media & Delivery

- **MongoDB Atlas** - Stores users, memories, sessions, settings, share links, subscriptions, and delivery records
- **Cloudinary** - Authenticated production media storage with AES-256-GCM encryption enabled by default
- **Web Push and VAPID** - Per-device opt-in delivery through provider endpoints and a browser service worker
- **Vercel and Render** - Frontend and API deployment with health checks and environment-based configuration

3. APPLICATION PAGES & COMPONENTS

Accounts & Guided Onboarding

Registration, sign-in, session restore, password recovery and change, profile setup, and a six-step guided tour introduce the main workflows.

Create & Edit Memories

Add a title, rich-text story, memory date, optional reminder, category, and up to 10 JPG, PNG, or WebP images of up to 8 MB each.

Optimized Image Pipeline

The browser and API create WebP full-view images and lightweight thumbnails; edits can retain, add, or remove individual images.

Timeline Home

Browse chronological cards with lazy paging, automatic loading, and timeline virtualization for larger desktop and mobile collections.

Calendar & Compact Tile Views

Switch to month-grouped calendar cards or a dense gallery, and save the preferred default view independently for each device profile.

Search, Sort, Filter & Favorites

Search titles, stories, and categories; sort newest or oldest; filter by category or favorites; and star important memories.

Rich Preview Carousel & Zoom

Open animated previews with multi-image navigation, swipe gestures, full-screen viewing, zoom and pan, image details, and direct actions.

3. APPLICATION COMPONENTS (CONTINUED)

Reminders & Browser Push

Set special dates and a lead window; review upcoming items; play an in-app sound; dismiss for today; snooze until tomorrow; and opt in to background push on each device.

Private Sharing & Public View

Create a time-limited public link for one memory, share through the device or clipboard, and present the owner, date, category, story, and images to recipients.

Image Export & Personal Backups

Export images from all memories, selected visible memories, or category/date/favorite filters; one image downloads directly and multiple images become a ZIP.

Hidden Images

Move items out of the main timeline into a dedicated interface protected by a four-digit UI PIN, then preview, unhide, or permanently delete them.

Trash & Recovery

Deleted memories remain recoverable for 30 days. Restore one or many items, permanently remove selected items, empty the bin, or allow automatic expiry.

Profile, Insights & Account Controls

Manage profile details and photo; view totals and a category chart; change the password; clear memories to Trash; revoke sessions; or delete the account.

Personalization, Backup & Accessibility

Separate desktop/mobile profiles cover themes, animated backgrounds, cards, controls, glass effects, fonts, hover, sounds, reminder lead time, reduced motion, and JSON settings backup/restore.

4. SECURITY, PRIVACY & DATA LIFECYCLE

Identity & Sessions

- **Strong password policy** - At least 10 characters with upper, lower, and numeric characters
- **bcryptjs hashing** - Passwords and Hidden PINs are stored as one-way hashes
- **Rotating sessions** - Short-lived JWT access plus opaque refresh tokens stored as hashes
- **Session revocation** - Logout, logout-all, password change, and reset invalidate sessions
- **Attempt controls** - Global API, login, reset, and Hidden PIN update throttles

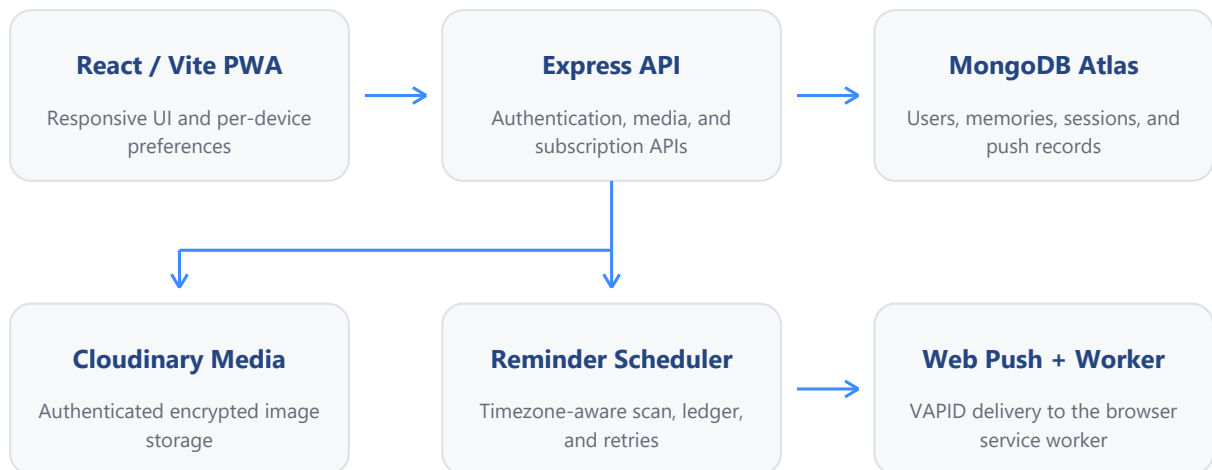
Request, API & Push Protection

- **CSRF and origin checks** - State-changing requests require a valid token and approved origin
- **CORS allowlist** - Credentialed browser access is limited to configured origins
- **Helmet policies** - Content Security Policy, frame blocking, and safer response headers
- **Ownership and sanitization** - Private media and queries are user-scoped; accepted rich content and settings keys are filtered
- **Push security boundary** - HTTPS provider allowlisting, key validation, VAPID signing, and private subscription fields

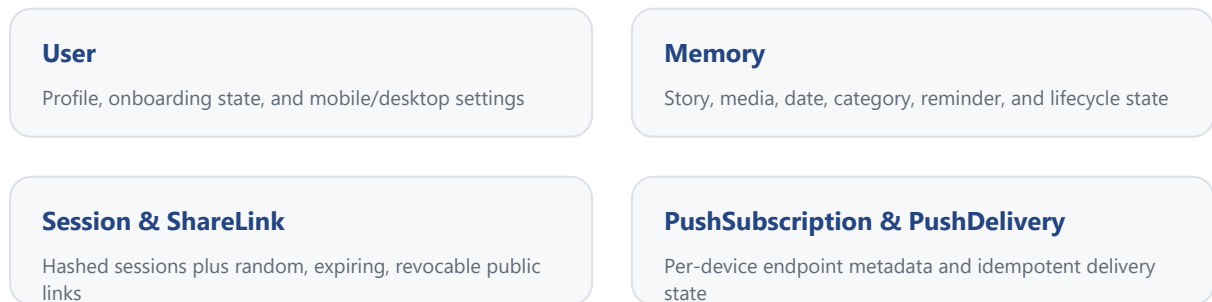
Media & Data Lifecycle

- **Strict upload validation** - MIME type, extension, count, file size, metadata, and pixel limits are checked
- **Normalized derivatives** - Sharp rotates, resizes, and writes full images and thumbnails as WebP
- **Protected delivery** - Cloudinary uses authenticated assets; private images stream through ownership-checked API routes
- **Encrypted cloud media** - AES-256-GCM is enabled by default before raw Cloudinary storage
- **Hidden Images PIN** - A salted PBKDF2-SHA-256 value gates the browser interface; the app password is required to change it
- **Expiring data paths** - Public links are time-limited, Trash purges after 30 days, and expired push endpoints and ledgers are cleaned up

5. SYSTEM ARCHITECTURE & DELIVERY



Core Data Models



Current Implementation Notes

- **Push opt-in** - Profile enables notifications separately on each browser or device; permission, service worker support, HTTPS, and stable VAPID keys are required
- **Private payload** - The notification contains a generic title and message, opens /timeline, and does not expose memory content on the lock screen
- **Delivery semantics** - A subscription/memory/date ledger provides idempotency, atomic leases, bounded retries, and retention; 404 or 410 responses remove stale endpoints
- **Time and scope** - IANA time zones and the user's lead window drive due scans; hidden, trashed, reserved, changed, and deleted reminders are excluded
- **Platform boundary** - On iPhone and iPad, push opt-in requires the app to be installed to the Home Screen and opened in standalone mode
- **Known product boundaries** - Visible export produces images or ZIP files; settings backup excludes memory records and photos; reset codes remain process-local without production delivery